

FAST

MOD  
123 DIV 10 = 3  
2  
1

123 = 3 \* 10 + 3  
123 = 3 \* 10 + 3

CU

123 DIV 10 num 12 sum 0  
= 12 123 3  
12 DIV 10 12 3  
12 DIV 10 1 3  
sum = 3

Date: \_\_\_\_\_

```

AS: 1. Declare num, sum ← INT
    2. Output "Enter number:"
    3. Input num
    4. While num DIV 10 != 0 Do
        Sum ← sum + (num MOD 10)
        num ← num DIV 10
    5. Endwhile
    6. Output sum
    7. num ← sum
    8. Endwhile
  
```

While num ≥ 10  
sum ← 0  
0 1  
0 2

output  
3  
11  
2

1. Declare sum and num as integer.
2. Initialise sum by 0 and Input num from user.
3. Use while loop <sup>until</sup> to check if the integer value of num divided by 10 is 0, meaning no more numbers left to add.
4. If not, take the mod of num by 10 and add it into sum.
5. Update num to by storing its integer division in it, so ~~the~~ addition can be possible.
6. ~~We~~ We would also need an outer loop incase the sum is more than 1 digit. So we will use another while to check if number is greater than 10.
7. Output sum each time inner loops end and store the sum into num. Outer loop will only run again if the sum is in two digit.